PROFESSIONAL PORTFOLIO

Godwin Etim Akpan

Geospatial & Data Engineering | Enterprise GIS | GeoAI | Cloud & Security

Selected projects in spatial intelligence, public-health analytics, secure data systems, climate and hazard exposure, and operational decision support

10+ YEARS	6,000+	400+	10+
GIS, analytics and spatial decision support	Maps, dashboards and analytical products	Professionals trained	Peer-reviewed publications

Portfolio Website • LinkedIn • godwinea.ai@gmail.com

Raleigh, North Carolina | Updated November 2026

PROFESSIONAL PROFILE

Turning complex data into trusted intelligence

I am a GIS, data analytics and data engineering professional with more than 10 years of experience transforming spatial, public-health and enterprise data into practical intelligence for informed decision-making.

My work connects data engineering, enterprise GIS, spatial analysis, GeoAI and secure analytical delivery. I build reproducible workflows and evaluated research prototypes, drawing on national public-health surveillance, emergency response and environmental GIS experience.

Professional evidence

- National-level public-health surveillance and emergency-response support in Liberia.
- Applied outbreak intelligence spanning COVID-19, malaria, measles, cholera, mpox, Ebola and Lassa fever.
- Enterprise GIS implementation, spatial data infrastructure, dashboard development and operational reporting.
- GIS training, capacity building, data-quality assurance, automation and applied spatial epidemiology.

CORE CAPABILITIES

Data Analytics
Python, SQL, dashboards, predictive analytics and business intelligence

Data Engineering
ETL/ELT, validation, feature engineering, PostgreSQL/PostGIS, Spark and PySpark

Enterprise GIS
ArcGIS Pro/Online, QGIS, spatial infrastructure, cartography and remote sensing

GeoAI & Machine Learning
XGBoost, Scikit-learn, spatial statistics, SHAP and responsible AI

Secure Systems
FastAPI, Docker, AWS, JWT, RBAC, audit logging and privacy-aware analytics

ACADEMIC FOUNDATION

MSc Big Data Technologies — All 180 credits completed and passed; final award pending

University of East London / UNICAF

Master of Technology in Space Science & Technology — Remote Sensing & Geoinformation Science

FUTA / ARCSSTE-E, Nigeria · Awarded October 2018

Geospatial delivery and analytical quality

Selected responsibilities connecting operational experience with current independent research and contract work.

Geospatial reviewer and AI evaluation

Independent contractor · September 2026–present

Review AI-generated geospatial workflows for methodological soundness, data provenance, spatial reasoning and compliance with project specifications. Client tasks and confidential evaluation material are not reproduced.

Geospatial benchmark development

Contract-program task · 2026

Built synthetic geospatial data-generation, validation and evaluation workflows. A registered 30-repetition pilot did not meet all predefined advancement criteria; development concluded and evidence was preserved.

Geoinformation Systems and Information Technology Specialist

AFENET · Full-time · Liberia · January 2021–January 2025

Administered enterprise GIS and national surveillance workflows; automated validation and recurring products; supported the WHO-hosted COVID-19 dashboard. Functional supervision included delegation, output review and mentoring. Produced 6,000+ maps, dashboards and products and trained 400+ professionals.

Geoinformation Specialist and Data Analyst

AFENET · Contract · Liberia · March–December 2020

Supported NPHIL and the Ministry of Health with harmonized surveillance datasets, initial COVID-19 dashboard preparation and daily geospatial reporting across all 15 counties.

Earlier GIS and outbreak-response work

Nigeria and Liberia · 2014–2020

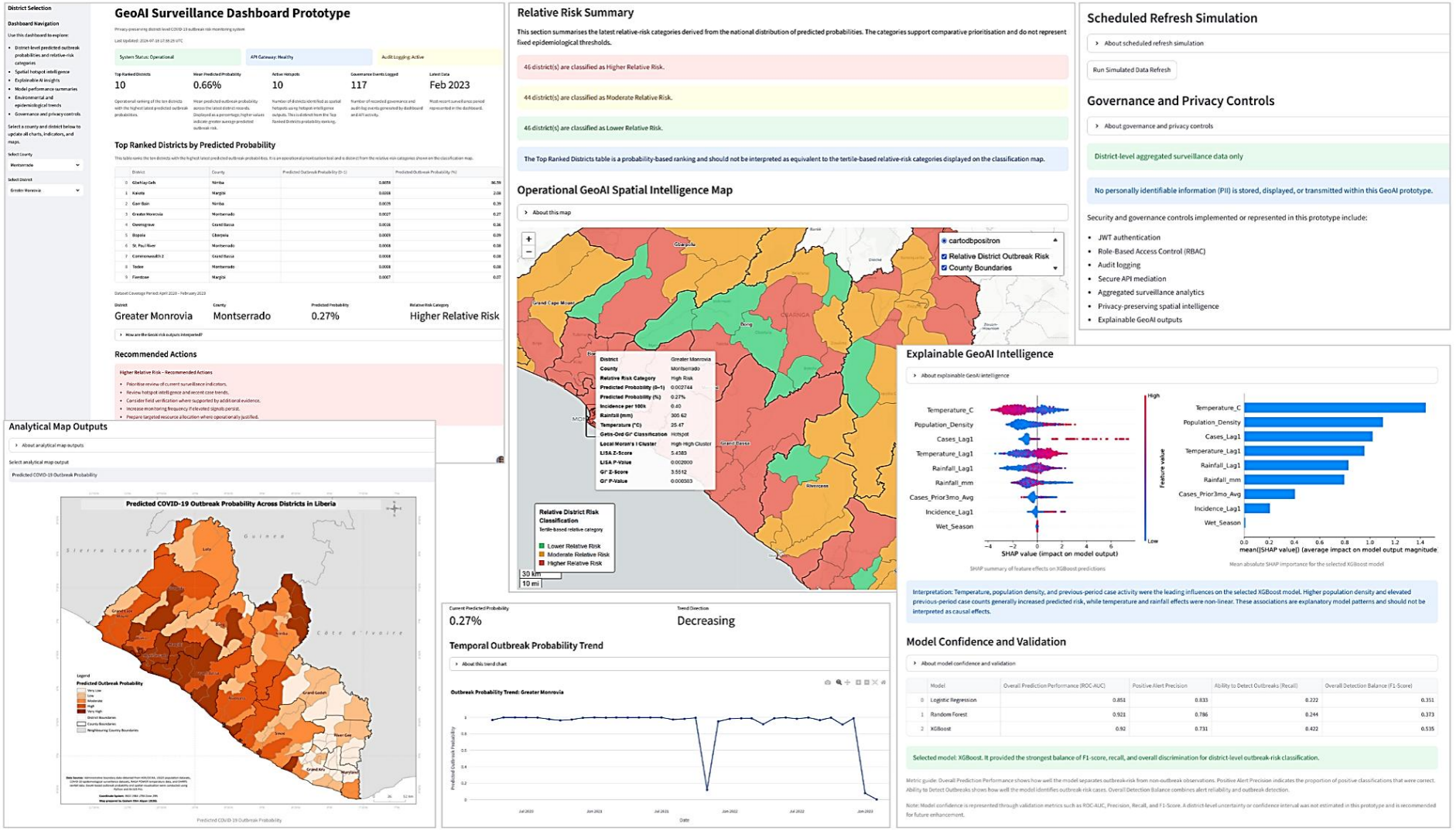
Environmental GIS, remote-sensing research, African Union Ebola-response deployment and Rivers State Ebola surveillance support established the operational foundation for later data and geospatial engineering.

View current experience and selected projects

FLAGSHIP PROJECT

Privacy-Preserving GeoAI Health Surveillance Dashboard

An integrated research prototype for risk ranking, spatial intelligence, explainability, governance monitoring and decision support.

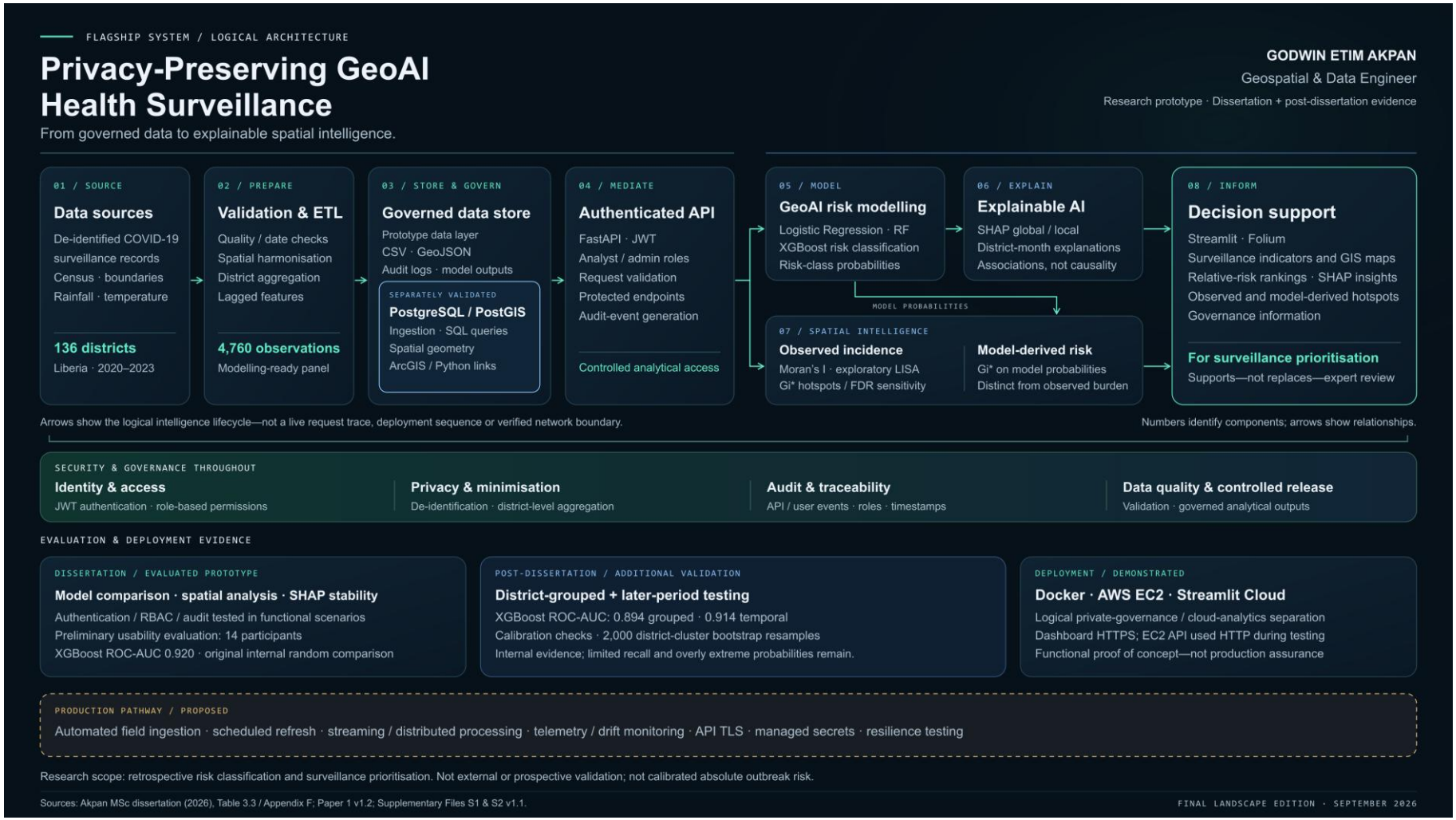


Integrated dashboard showing risk rankings, interactive mapping, SHAP explanations, governance controls and model validation.

Research-prototype boundary: The public demonstration uses aggregated, de-identified district-level data and is not intended for clinical diagnosis or operational deployment without further validation, security hardening, governance review and institutional approval.

GeoAI health surveillance architecture

Implemented analytics and access controls, separately validated PostGIS, demonstrated deployment and proposed production capabilities are identified explicitly.

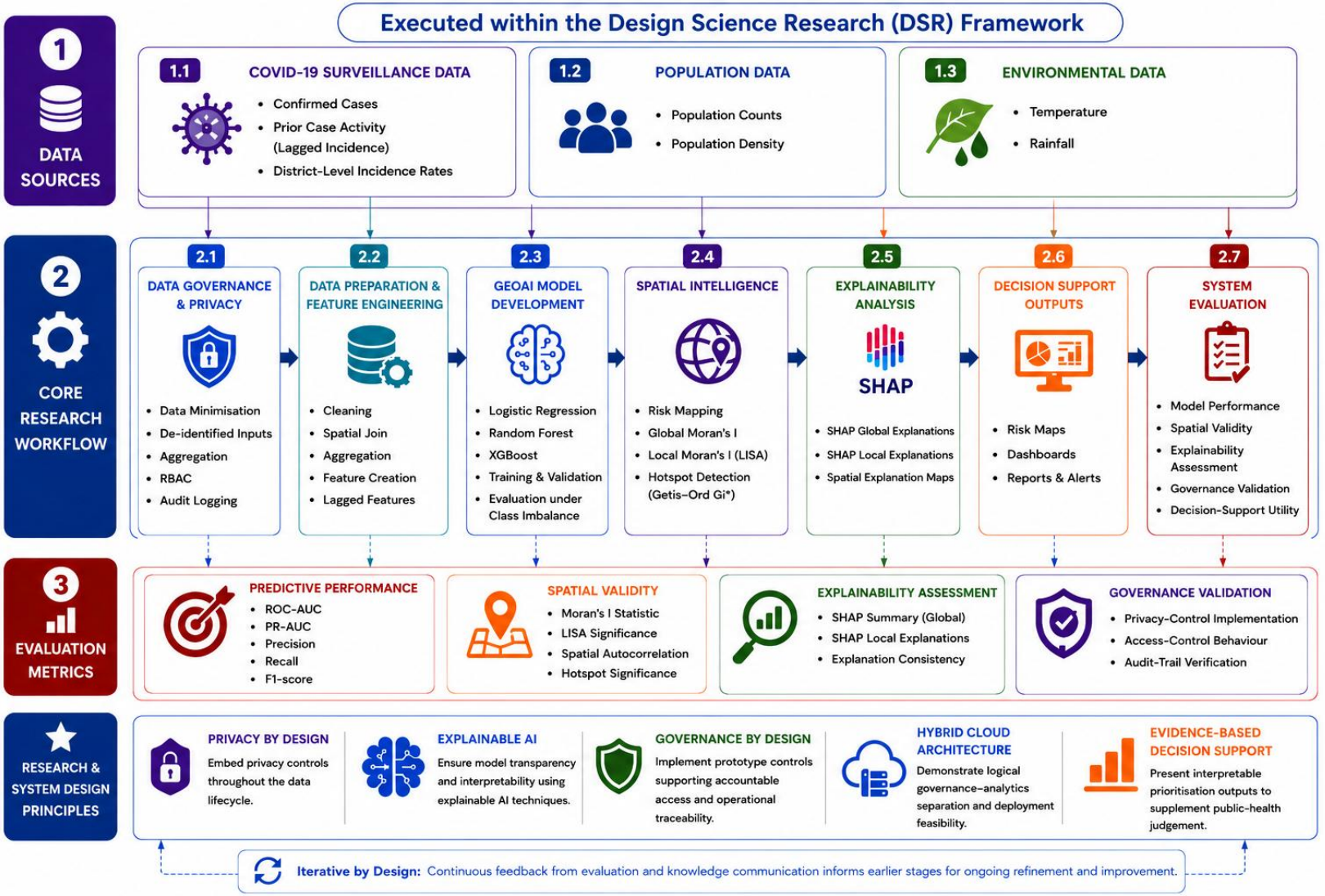


Current logical architecture used in the engineering case study. View the full-size diagram and evaluation evidence on the website.

Implementation scope: CSV/GeoJSON data preparation, spatial analytics, model evaluation, SHAP, Streamlit/Folium, FastAPI, JWT, RBAC and audit-event generation were implemented. PostGIS was validated separately. Production network isolation, scalable ingestion and advanced operations remain proposed extensions.

Design Science Research Workflow

The dissertation evaluates the artefact as an integrated research and engineering system—not simply as a predictive model.



Research workflow spanning data sources, governance, feature engineering, GeoAI, spatial intelligence, explainability and evaluation.

Evaluation dimensions: Predictive performance · spatial validity · explainability · security and governance · usability · decision-support value.

Validation across geography and time

The original dissertation comparison and subsequent internal validation answer different questions.

Evidence stage	Evaluation design	ROC AUC	Recall	F1
Original dissertation	Random district-month comparison	0.920	0.422	0.535
Post-dissertation	Five-fold district-grouped validation	0.894	0.200	0.291
Post-dissertation	15-month later historical holdout	0.914	0.326	0.394

XGBoost results. The dataset contains 136 districts and 4,760 modelling observations from the same historical surveillance setting. The later tests are not external or prospective validation.

Model comparison

XGBoost had the strongest recall/F1 balance in the original comparison. Logistic Regression achieved higher F1 in both subsequent designs; XGBoost retained the highest ROC AUC.

Calibration and uncertainty

XGBoost recalibration slopes of 0.498 (grouped) and 0.525 (temporal) indicated overly extreme probabilities. The primary temporal holdout contained only 43 positives; classification intervals were wide.

Spatial and user evidence

Five of ten unadjusted observed-incidence hotspots remained after false-discovery-rate correction. Fourteen participants supplied preliminary usability feedback; the later dashboard refinement was not formally re-evaluated.

Engineering boundary

Functional tests demonstrated authentication, role restrictions and audit events. They did not establish penetration-test assurance, tamper-proof logs or production resilience. The prototype API used HTTP during testing; production API TLS remains required.

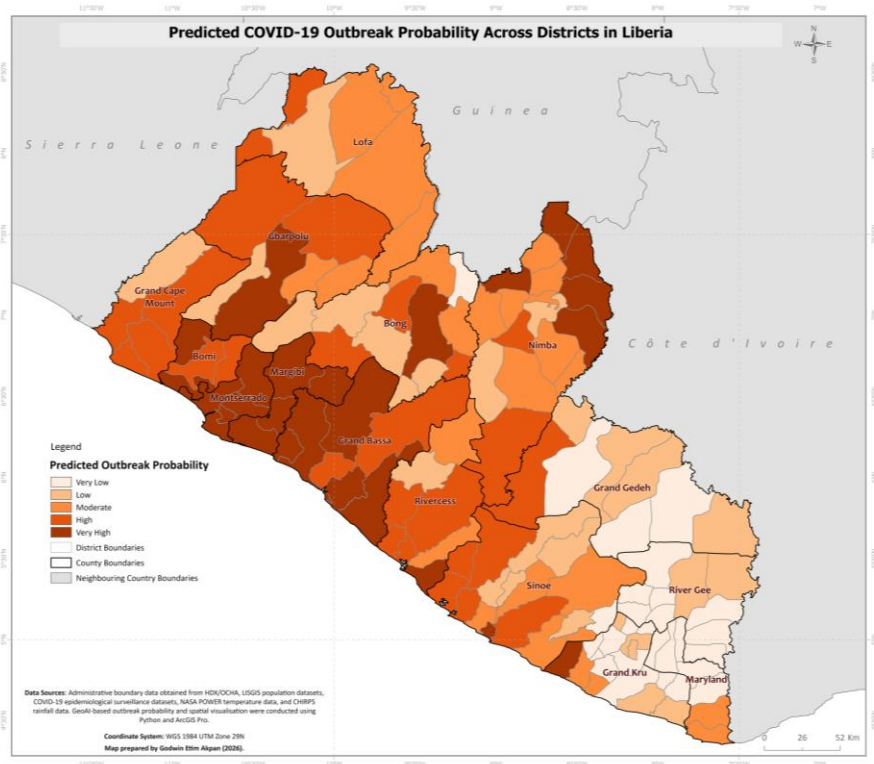
Evidence: dissertation model comparison; subsequent coauthor-review manuscript v1.2, Table 7, and Supplementary File S2 v1.1, as documented in the engineering case study. This manuscript is not represented as a published peer-reviewed article.

Inspect architecture and evaluation evidence

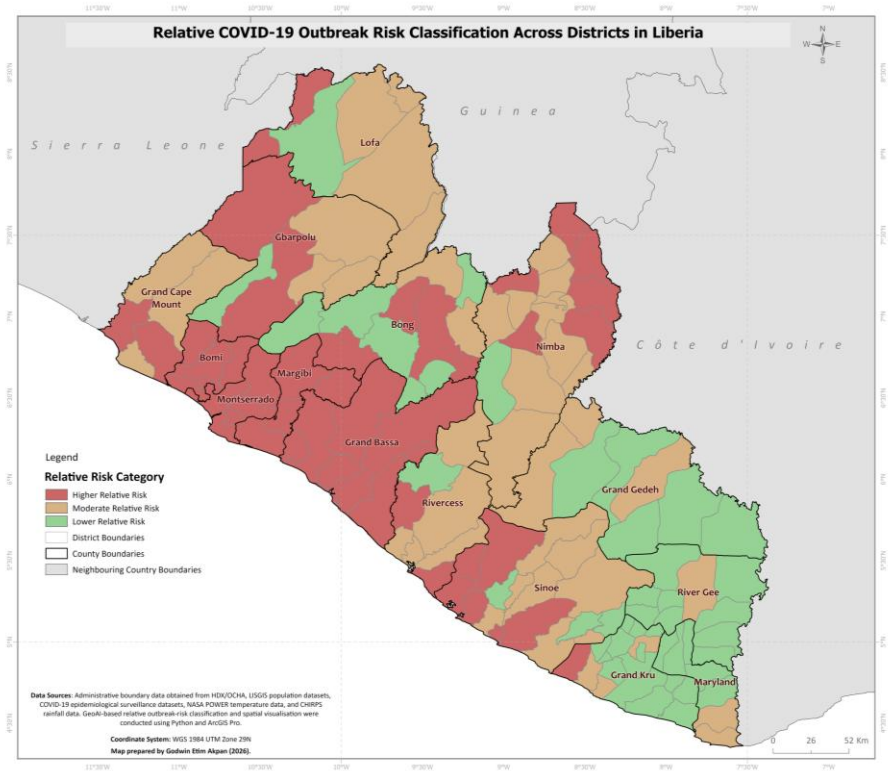
From Predicted Probability to Decision-Support Classification

Probability supports analytical ranking; relative-risk categories provide an interpretable comparative view for prioritisation.

Predicted outbreak probability



Relative district-risk category

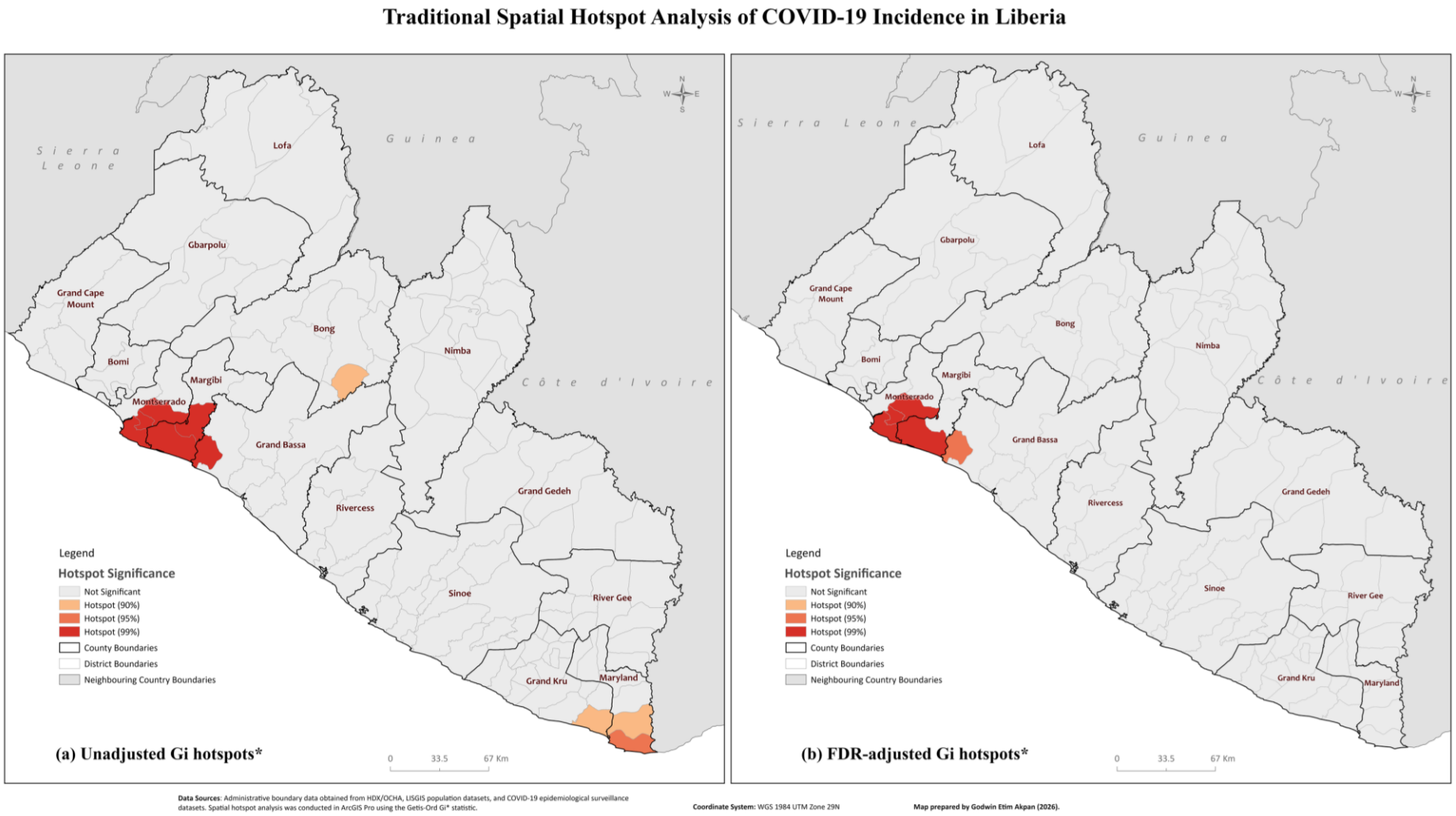


District-level demonstration outputs derived from the dissertation's XGBoost-based GeoAI workflow.

Interpretation: The maps communicate comparative risk and model output; they do not represent fixed epidemiological thresholds or clinical predictions.

Hotspot Sensitivity and Multiple-Testing Adjustment

Comparing original Getis-Ord Gi* classifications with false-discovery-rate-adjusted results supports more defensible spatial interpretation.



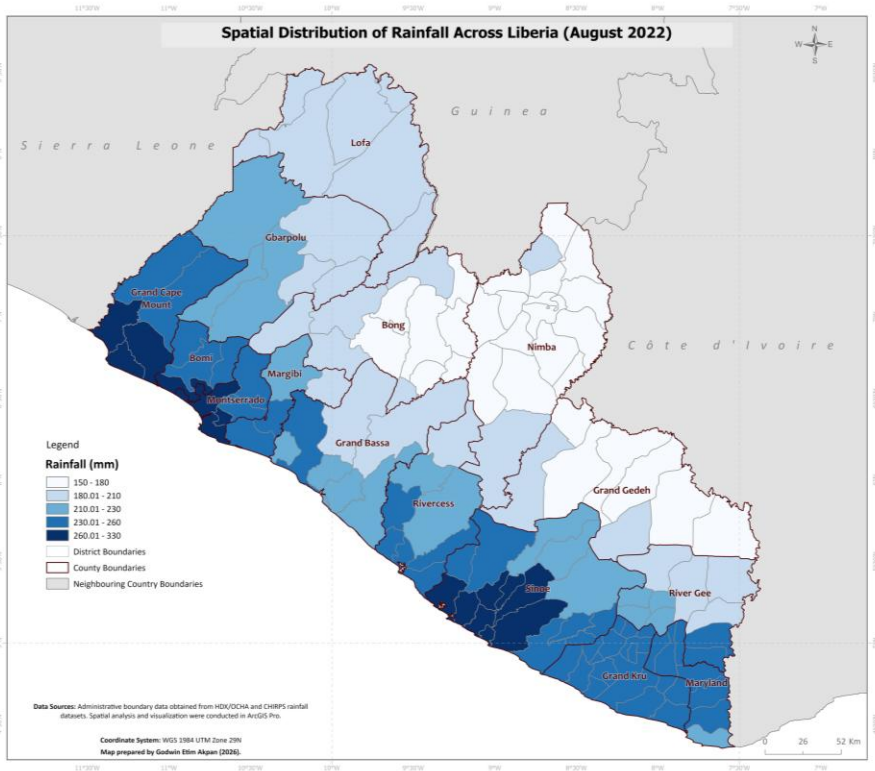
Traditional hotspot analysis of COVID-19 incidence: unadjusted Gi results compared with FDR-adjusted results.*

Analytical value: This comparison demonstrates awareness that apparently significant spatial patterns can change after correction for multiple testing—an important safeguard against overinterpretation.

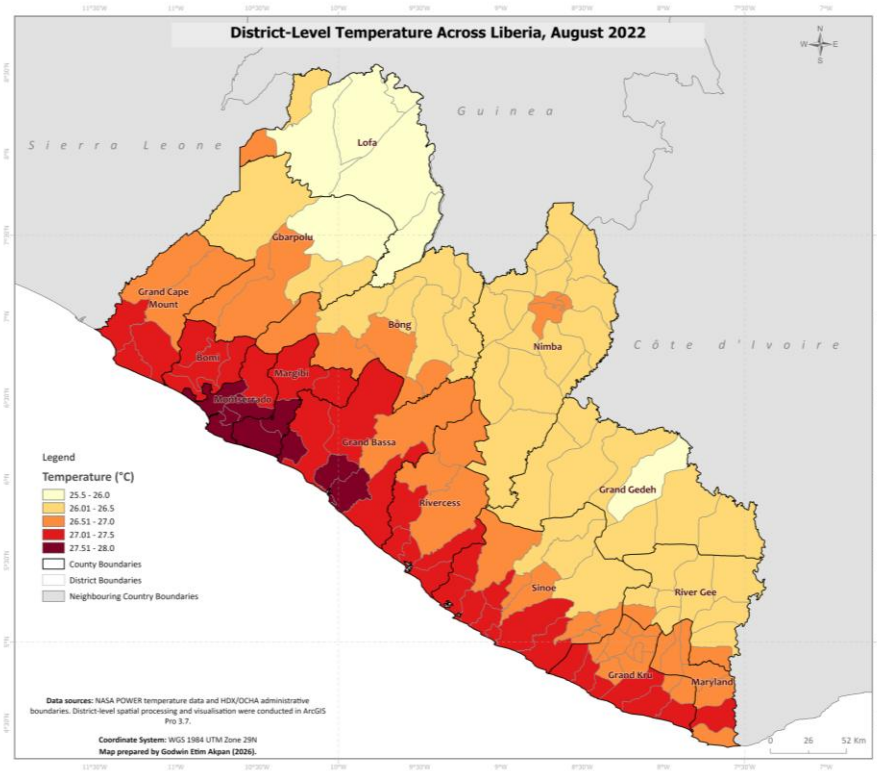
Rainfall and Temperature as Spatial Risk Features

Environmental indicators were spatially joined to district-level surveillance and population data for exploratory modelling and explanation.

Rainfall distribution — August 2022



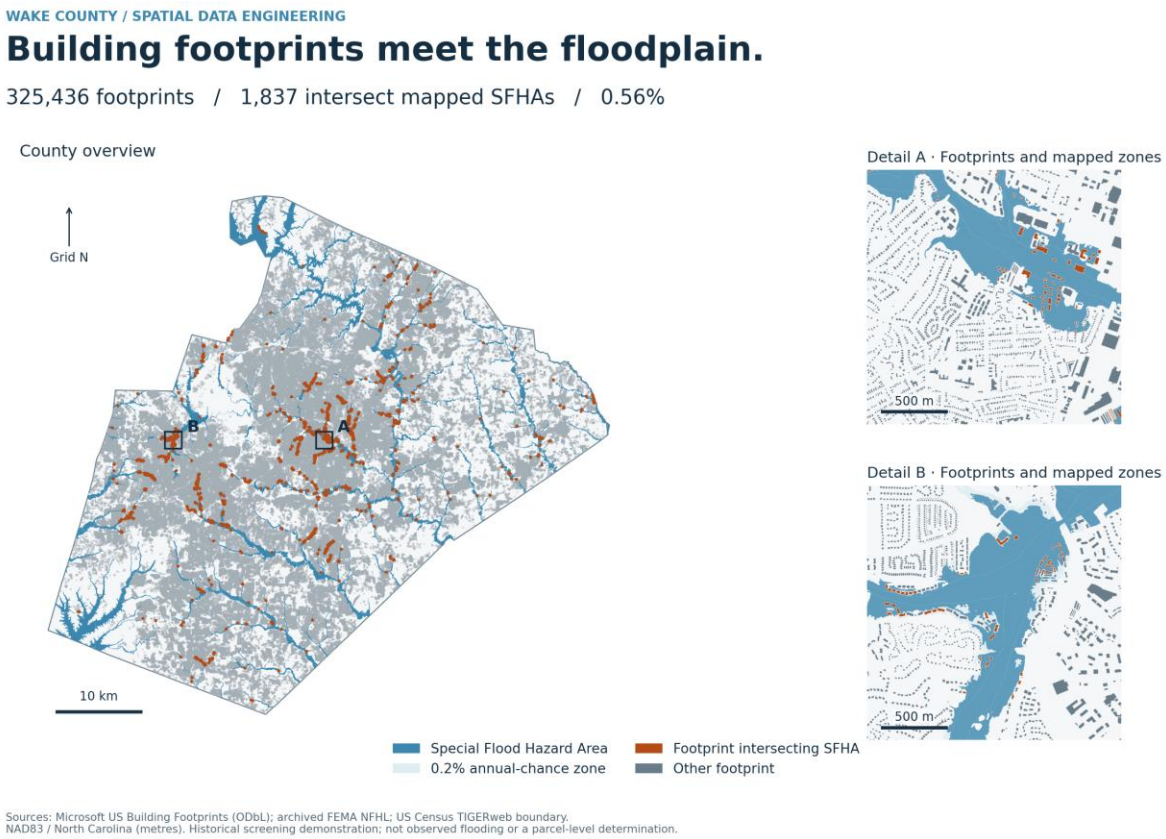
District-level temperature — August 2022



Environmental data: CHIRPS rainfall and NASA POWER temperature; administrative boundaries from HDX/OCHA and national sources.

Wake County building exposure and spatial QA

Validated footprint processing and exposure analysis using Microsoft building footprints, archived FEMA flood polygons and a Census county boundary. Corrected analysis: October 2026.



325,436 distinct county footprints

1,837 intersect mapped Special Flood Hazard Areas

0.564% of the county inventory

WORKFLOW

Processed the 4.68 million-footprint statewide source; assigned county membership consistently; repaired geometry; checked duplicates; retained small footprints with quality flags.

VALIDATION

Reconciled saved outputs and exposure flags against a dissolved floodplain. Independently checked intersection areas for 100 sampled buildings.

SCOPE

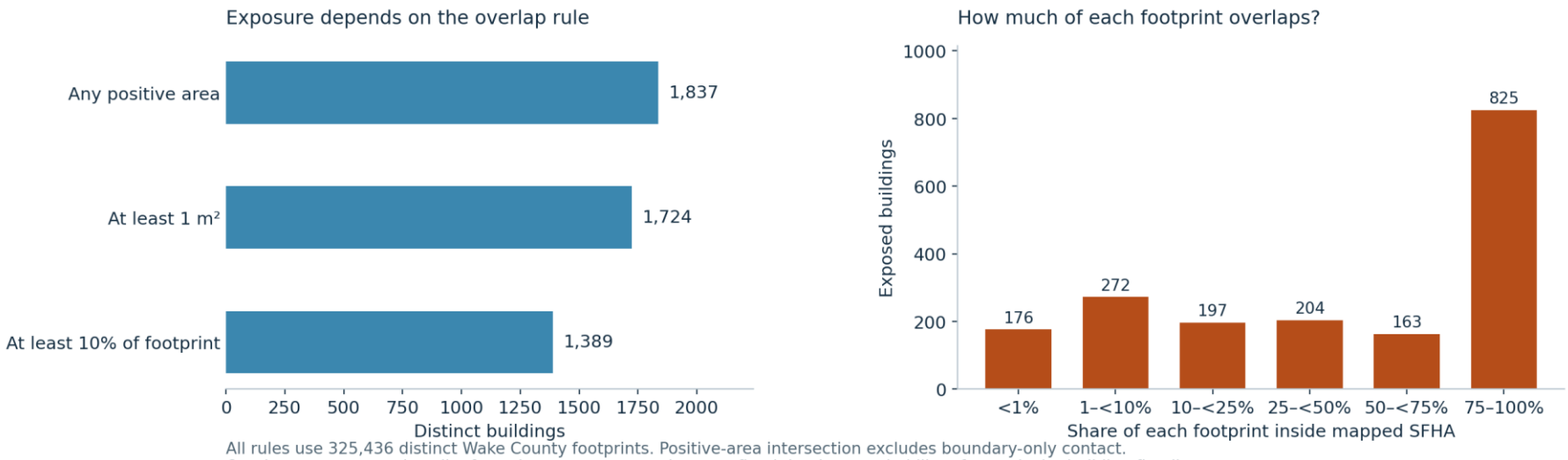
Historical geometric exposure, not observed flooding, damage or present-day hazard. Independent portfolio demonstration; not commissioned by NCEM.

Exposure definitions and independent checks

The same inventory supports three explicit overlap thresholds; distinct-building counts avoid duplicate intersection rows.

METHOD SENSITIVITY / SAME COUNTY DENOMINATOR

A count with a definition you can inspect.



All rules use 325,436 distinct Wake County footprints. Positive-area intersection excludes boundary-only contact.
Overlap percentages describe footprint geometry, not damage, flood depth, or probability of a particular building flooding.

Exposure definition	Distinct buildings	Share of 325,436
Any positive intersection area	1,837	0.564%
At least 1 square metre	1,724	0.530%
At least 10% of footprint area	1,389	0.427%

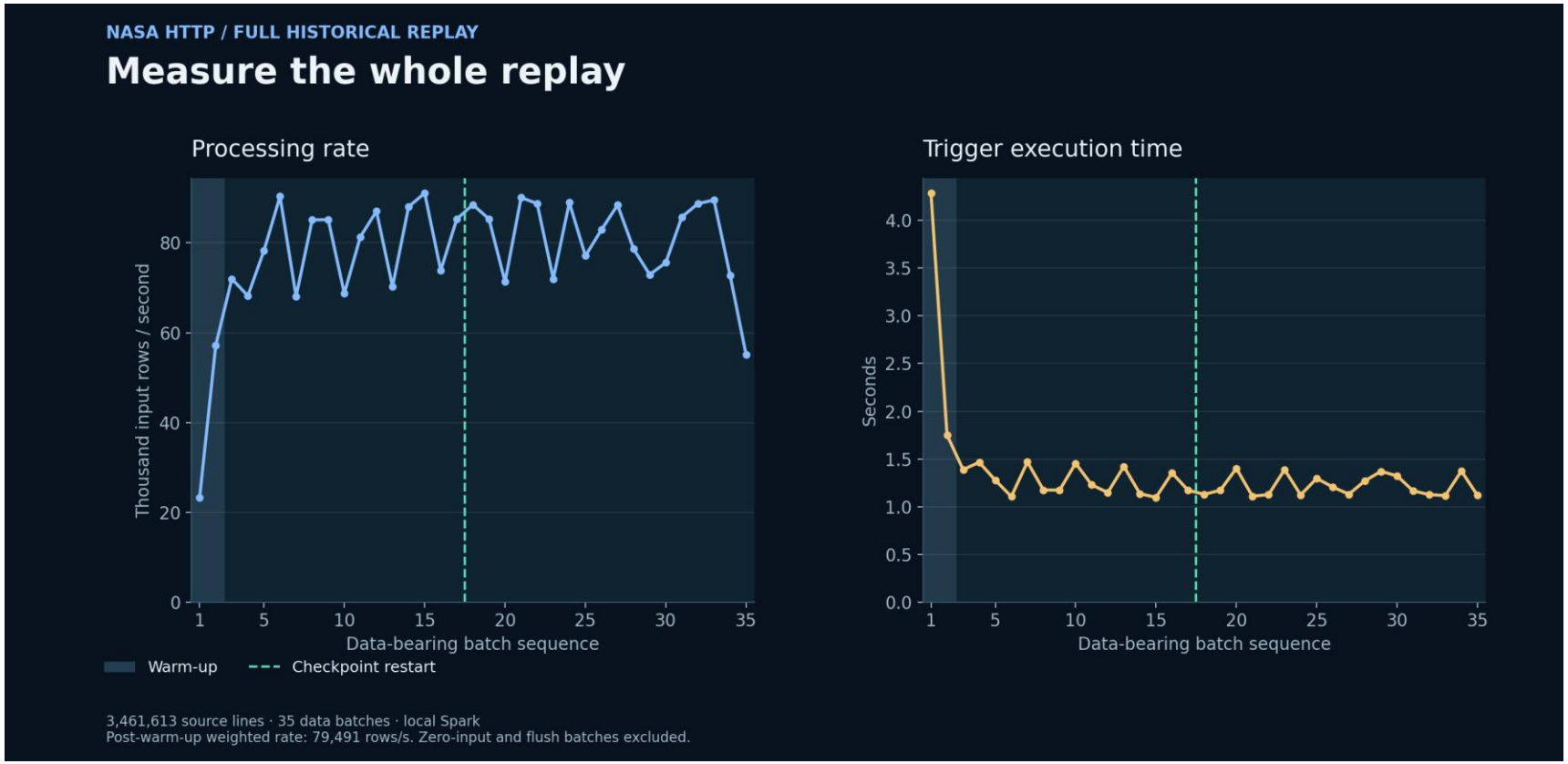
Validation passed checks on all 325,436 saved buildings, including unique identifiers, county membership, metric units, valid geometry and dissolved-hazard exposure identity. Intersection areas were independently recalculated for a 100-building sample.

Explore the Wake County case study and result downloads

NASA HTTP log streaming

Spark Structured Streaming with event-time processing, strict parsing and checkpoint recovery.

3,461,612 valid requests · 35 data batches · 183,308 final minute/status windows



Full historical replay trace showing processing behavior, warm-up and checkpoint restart.

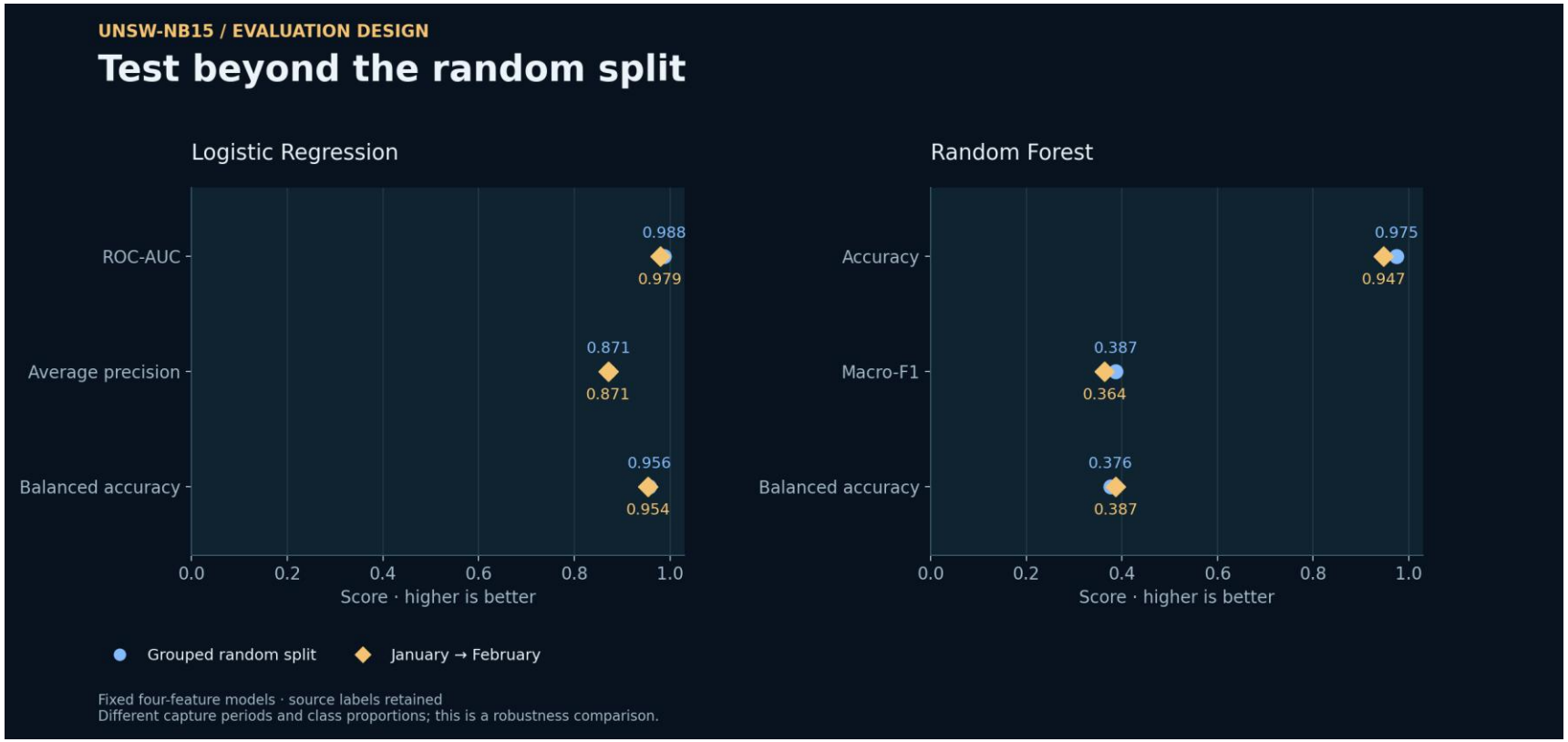
Built a reproducible replay pipeline, retained malformed-row outcomes and batch metrics, and reconciled every final window against an independent parser. Synthetic contracts test late events and recovery. This is an ordered historical replay, not a live NASA source.

Case study, implementation and evidence

Distributed security analytics

PySpark classification across 2.54 million UNSW-NB15 network flows.

Later-capture binary ROC AUC 0.979 · Separate ten-class evaluation



Grouped-random and later-capture evaluation reveal how performance changes with the test cohort.

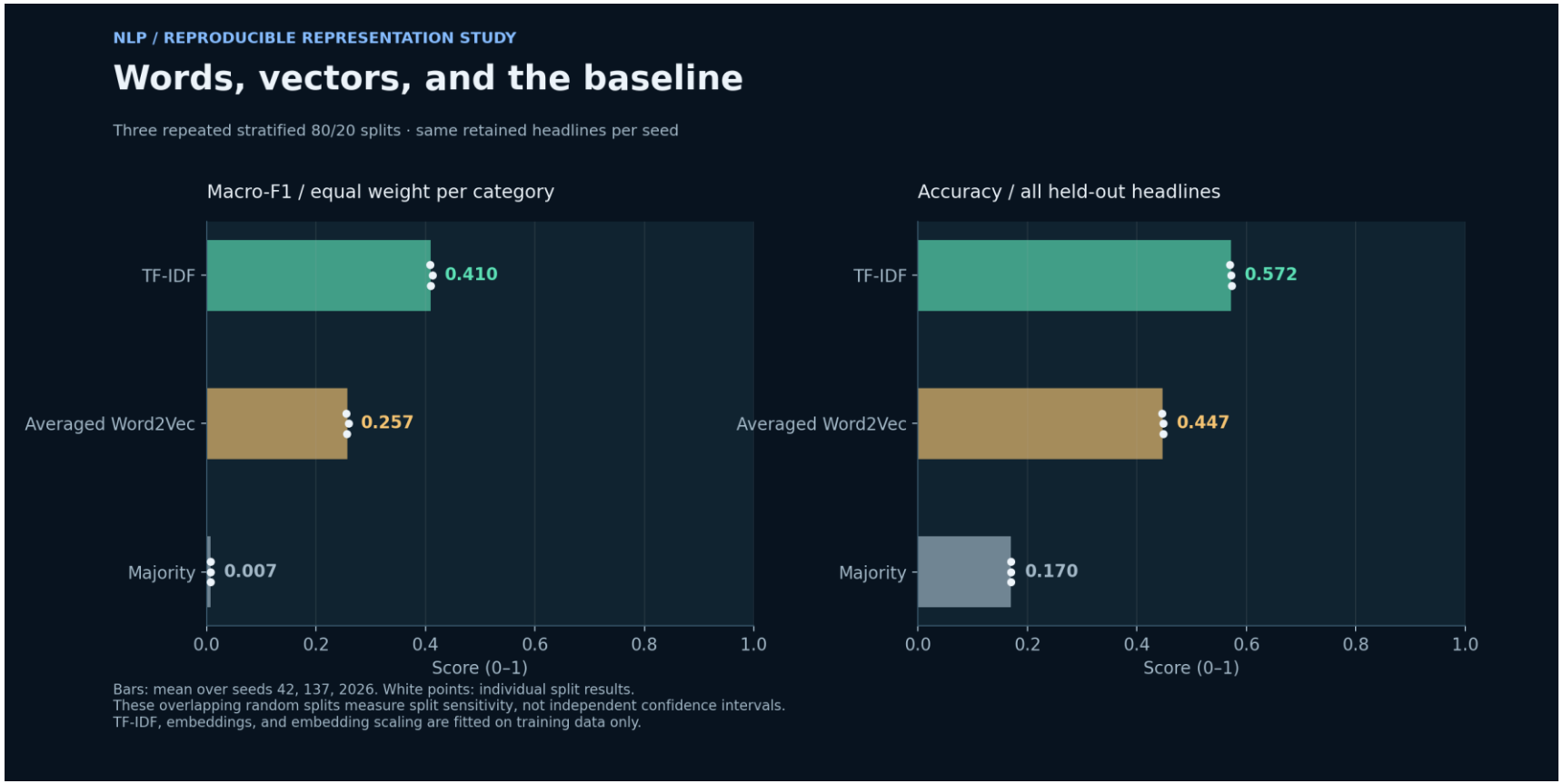
Validated class-label mapping, used duplicate-aware partitions and retained named per-class results. Strong binary discrimination does not imply equally strong multiclass detection; the later-capture Random Forest macro-F1 was 0.364.

Case study, implementation and evidence

NLP and document clustering

Reproducible text ingestion, feature learning, classification and held-out topic evaluation.

206,959 headlines · 42 categories · Mean TF-IDF macro-F1 0.410



Representation comparison across three declared splits; macro-F1 gives each category equal weight.

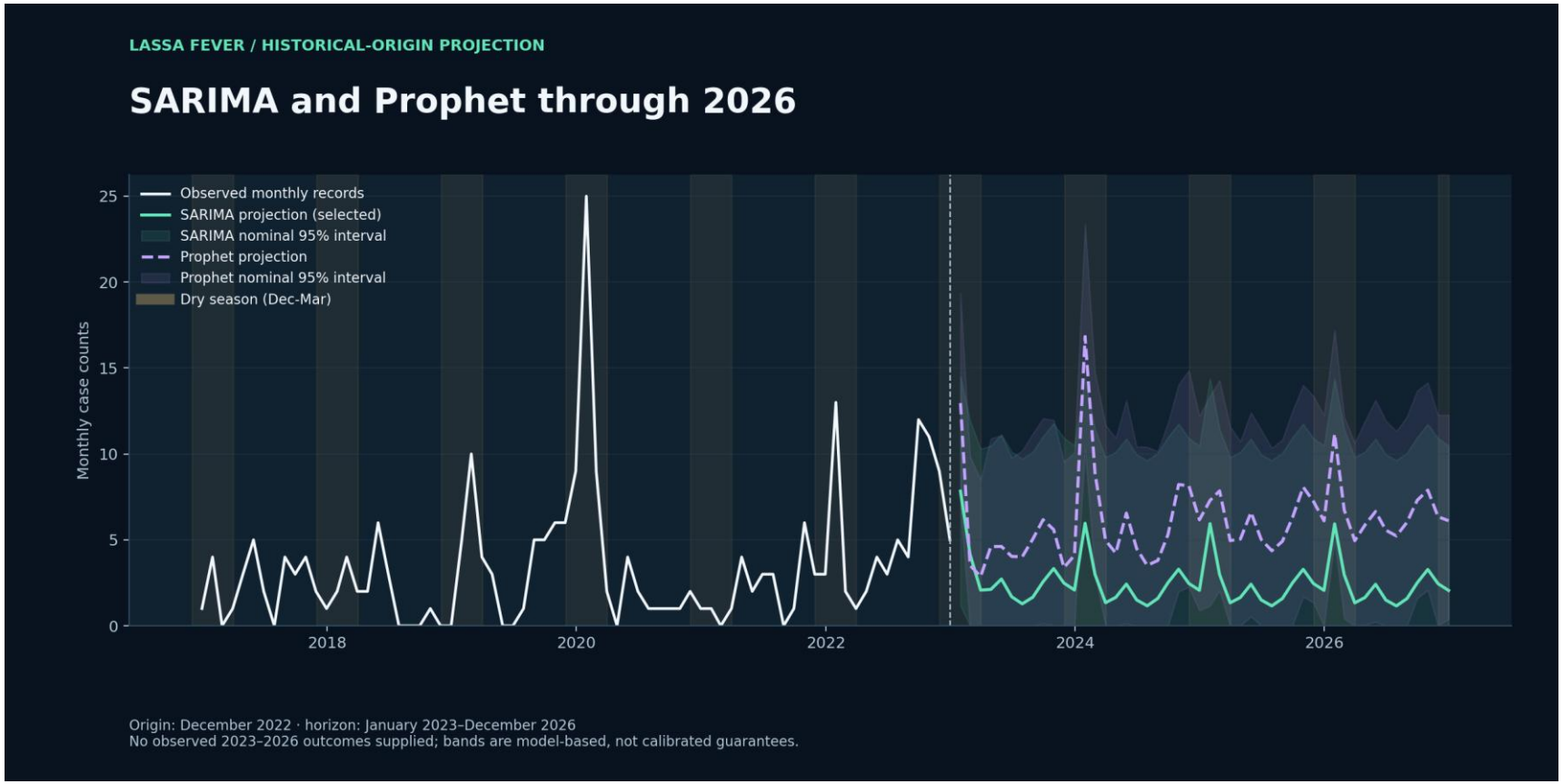
Compared TF-IDF and averaged Word2Vec with Logistic Regression, fitted transformations on training data and retained per-category evaluation. TF-IDF averaged 57.2% accuracy; averaged Word2Vec macro-F1 was 0.257 under the declared settings. A linked experiment evaluates rare-word probability estimation.

Case study, implementation and evidence

Lassa fever surveillance forecasting

Monthly panel preparation, national rolling-origin forecasting and county benchmarking for Liberia.

15 counties · 73 observed months · Three annual backtests · SARIMA national MAE 3.04



Observed records end in December 2022. SARIMA and Prophet projections run through 2026; shading marks December–March.

Retained counties with no recorded outbreaks as historical zeros. The county zero baseline had the lowest MAE (0.272), while recent mean had the lowest RMSE. Nominal model intervals and historical-origin projections are not guarantees or validated current forecasts.

Case study, implementation and evidence

OPERATIONAL DELIVERY

Emergency GIS Data Engineering and QA/QC

Trusted decision-support products begin with controlled ingestion, validated attributes, spatial consistency and reproducible publishing.

1	2	3	4	5
UNDERSTAND	**PREPARE**	**INTEGRATE**	**DELIVER**	**SUSTAIN**
Clarify operational users, decisions, timing, data owners and minimum viable outputs.	Validate schema, clean attributes, standardise coordinate systems and resolve geometry issues.	Join spatial and non-spatial data, engineer features and document lineage and assumptions.	Publish maps, dashboards, reports and APIs with clear interpretation and responsible-use notes.	Monitor refreshes, troubleshoot data quality, maintain documentation and support users.

Selected evidence

- Digitised and validated Liberia's road network using field-collected GPS observations and satellite imagery.
- Standardised road class, condition and accessibility attributes for integration into national GIS datasets.
- Applied validation, cleaning, aggregation, feature engineering and spatial joins to prepare model-ready surveillance data.
- Delivered time-sensitive GIS products supporting outbreak response, planning and operational reporting.

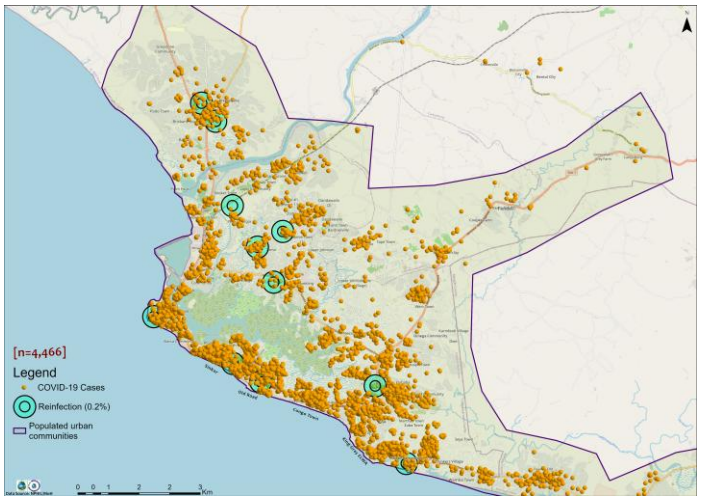
Engineering stack

Python · SQL · Pandas · GeoPandas · PostgreSQL/PostGIS · ETL/ELT · data validation · feature engineering · ArcGIS Pro · QGIS · FastAPI · Docker · Git/GitHub

Delivery principle

Every analytical result should be traceable to its data, assumptions, validation checks and intended decision context.

Emergency GIS and surveillance reporting

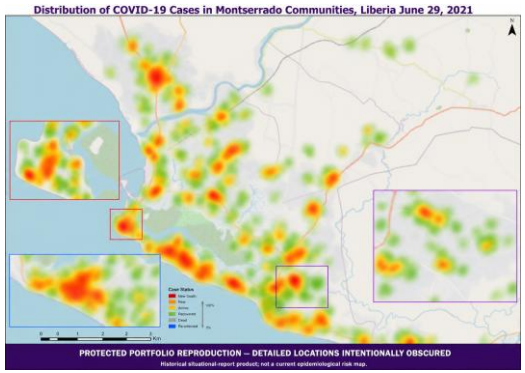


Published spatial epidemiology research: Spatial distribution and clustering of COVID-19 cases in Liberia. Reproduced from Akpan et al. (2022), *COVID-19 Reinfection in Liberia: Implication for Improving Disease Surveillance*, PLOS ONE, 17(3), e0265768. https://doi.org/10.1371/journal.pone.0265768. Licensed under CC BY 4.0.

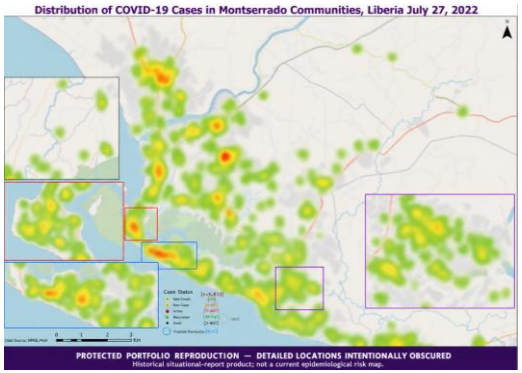
Supported national Ebola and COVID-19 emergency-response operations by producing daily and on-demand GIS products that integrated evolving surveillance and geographic information for situational awareness, resource prioritisation and partner coordination.

- Delivered more than 6,000 maps, dashboards and analytical products.
- Supported field data collection, troubleshooting, reporting and stakeholder communication.
- Trained more than 400 professionals in GIS, spatial analysis and digital data collection.
- Developed validated time-series workflows comparing surveillance patterns across June 2021, July 2022 and February 2023.
- Applied consistent symbology, spatial validation and descriptive analysis without presenting causal or formal hotspot conclusions.

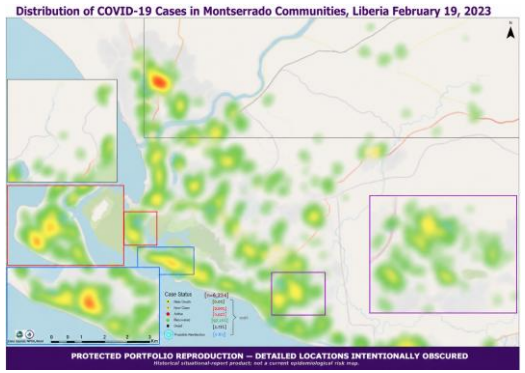
COVID-19 Case Distribution — June 2021



COVID-19 Case Distribution — July 2022



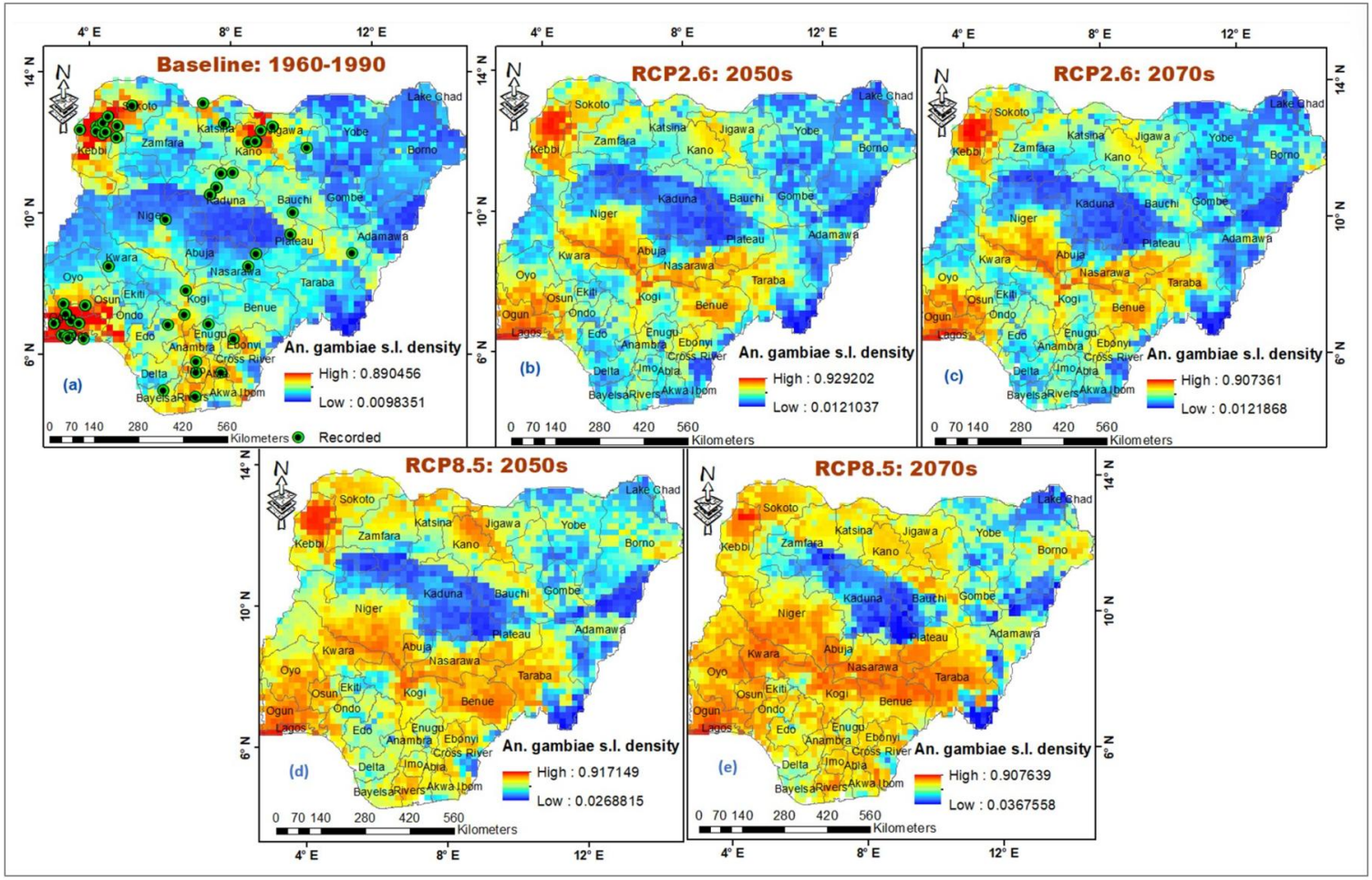
COVID-19 Case Distribution — February 2023



Historical operational maps previously disseminated through authorised channels. Detailed locations are obscured. These panels demonstrate surveillance reporting and cartographic workflows, not current epidemiological risk.

Potential Distribution of Malaria Vectors Under Climate Scenarios

A MaxEnt-based spatial modelling project examining how environmental suitability for Anopheles mosquitoes could shift under projected climate conditions in Nigeria.



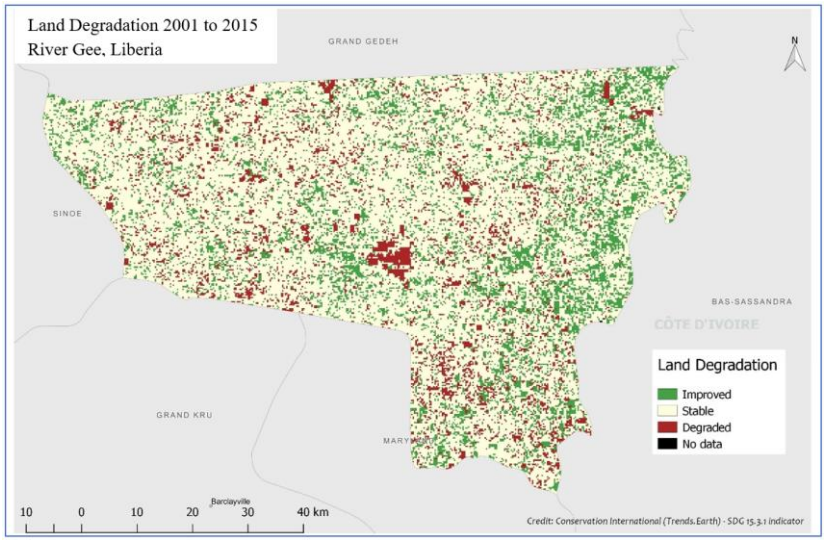
Baseline and projected suitability scenarios produced using MaxEnt and GIS-based bioclimatic modelling.

Capabilities demonstrated: Species-distribution modelling · remote sensing · climate scenarios · environmental raster processing · spatial interpretation.

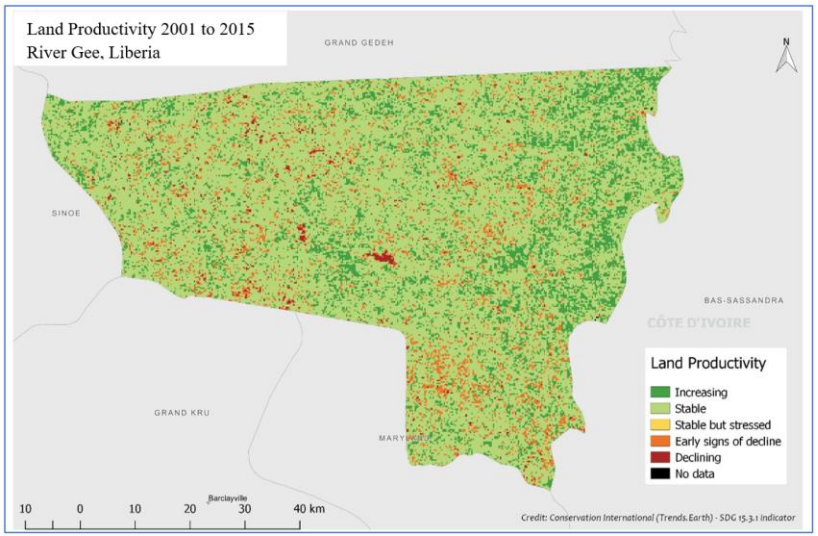
Land Degradation and Productivity Assessment — River Gee County

An SDG 15.3 pilot project assessing land-productivity trends, land-cover dynamics and environmental change from 2001 to 2015.

Land degradation



Land productivity



PROJECT CONTRIBUTION

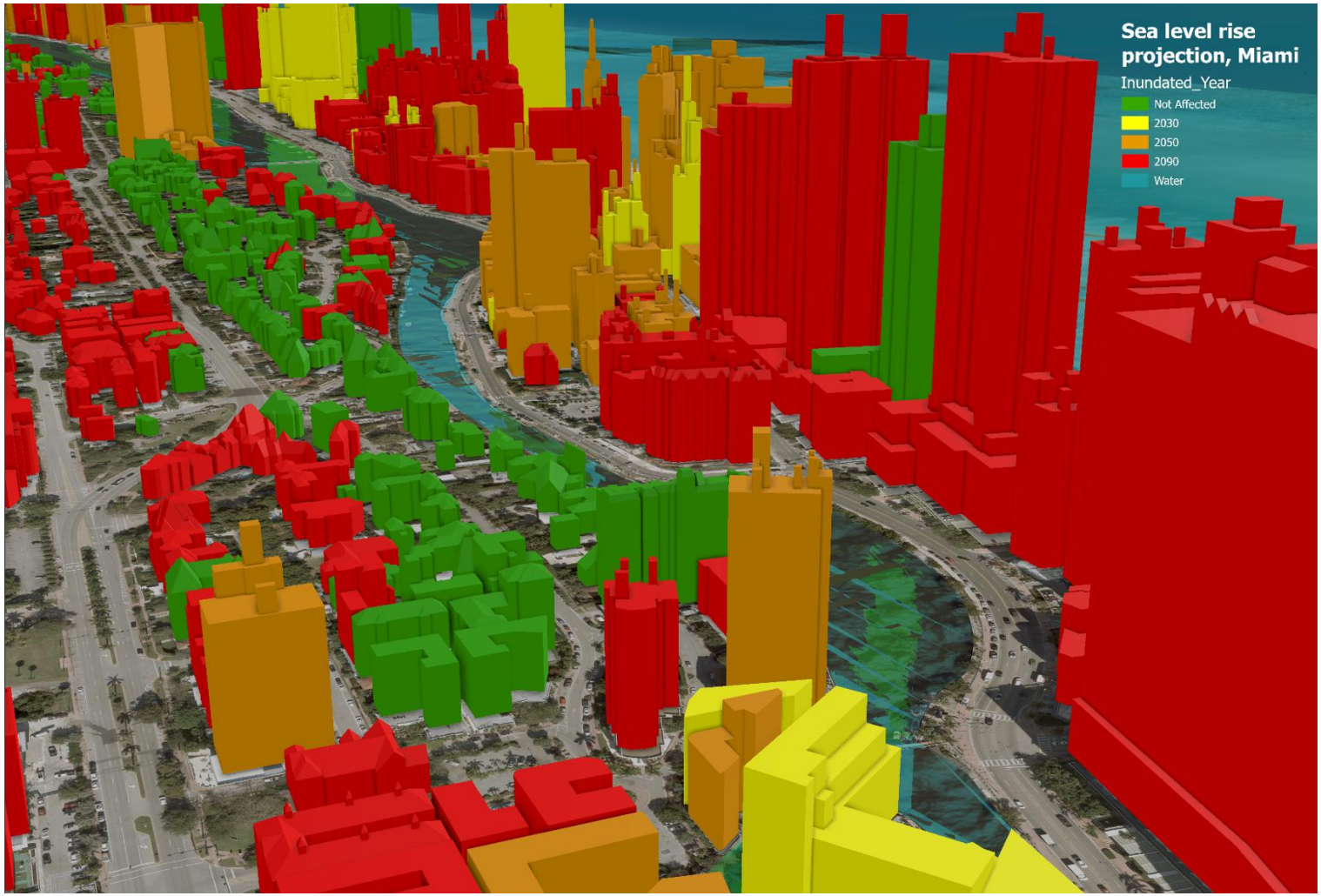
- Designed and executed an independent SDG-monitoring proof of concept.
- Processed productivity, land-cover and soil-organic-carbon indicators.
- Mapped land-degradation and productivity patterns at high resolution.
- Translated technical results into stakeholder-oriented environmental intelligence.

TOOLS

Trends.Earth · ArcGIS · remote sensing · spatial analysis

Miami sea level rise visualization

A 3D GIS exercise illustrating how scenario-based symbology communicates potential urban exposure.



Illustrative scenario visualization. The years and colors reflect the source exercise; they are not validated building-level inundation forecasts.

Analysis

Elevation and land-cover data prepared for scenario-based visualisation.

Communication

3D extrusion and colour communicate relative exposure and changing risk.

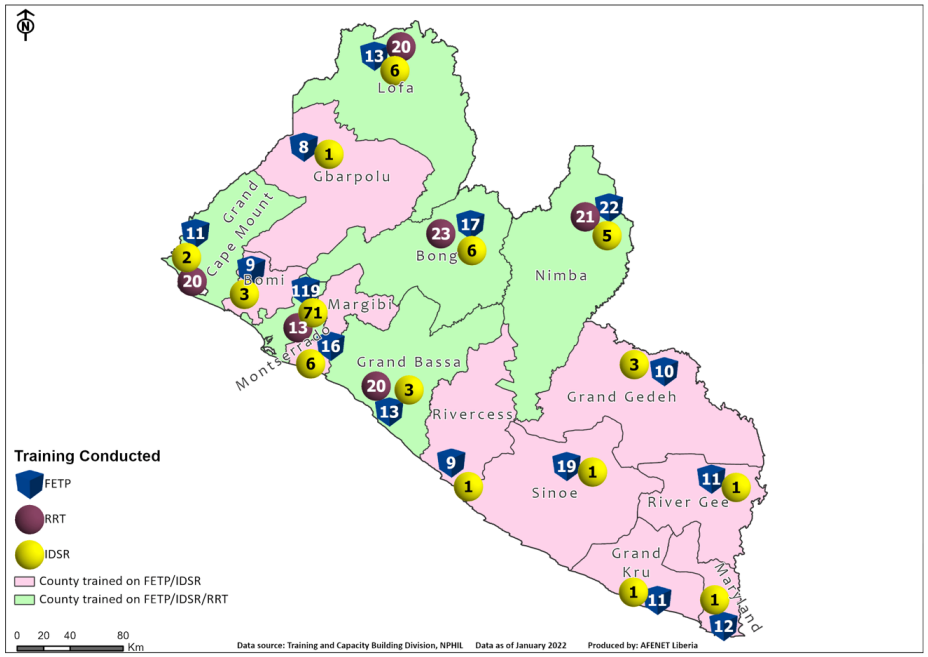
Application

Demonstrates 3D GIS communication; scenario inputs and local exposure require independent validation before planning use.

Workforce Coverage and Cross-Border Surveillance

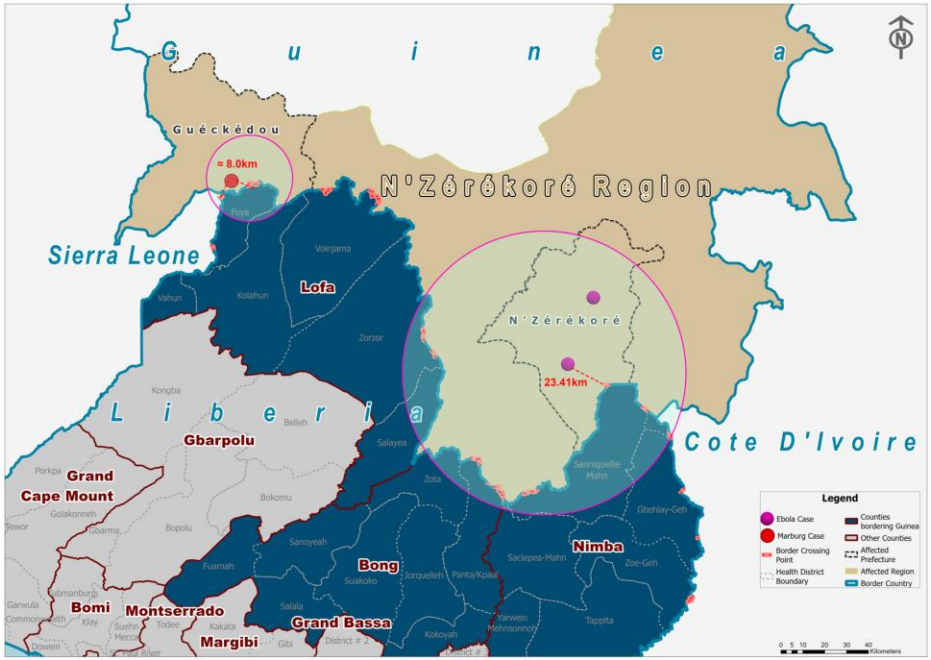
Operational cartography supporting public-health capacity building, incident preparedness and geographically targeted surveillance.

Emergency preparedness training coverage



Visualises national workforce-development reach and identifies counties with limited or no documented coverage.

Ebola and Marburg cross-border mapping



Integrates incident proximity, administrative boundaries and border context to support preparedness and surveillance prioritisation.

Selected operational public-health mapping undertaken in Liberia with institutional partners.

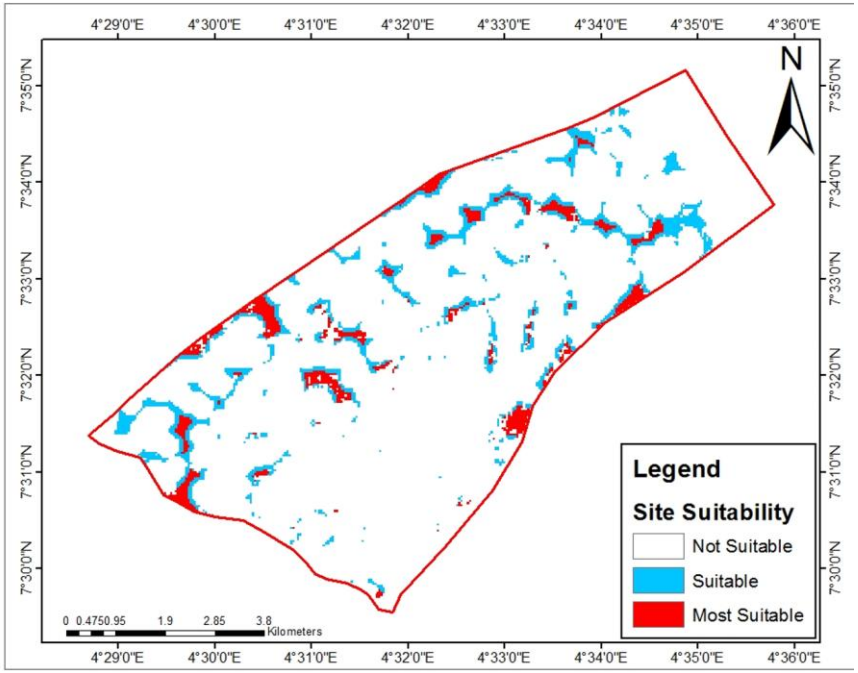
Publication note: Only aggregated or publicly releasable content should be included in externally distributed versions; organisational attribution and permissions should be retained.

INFRASTRUCTURE PLANNING

Airport Site-Suitability Modelling — Obafemi Awolowo University

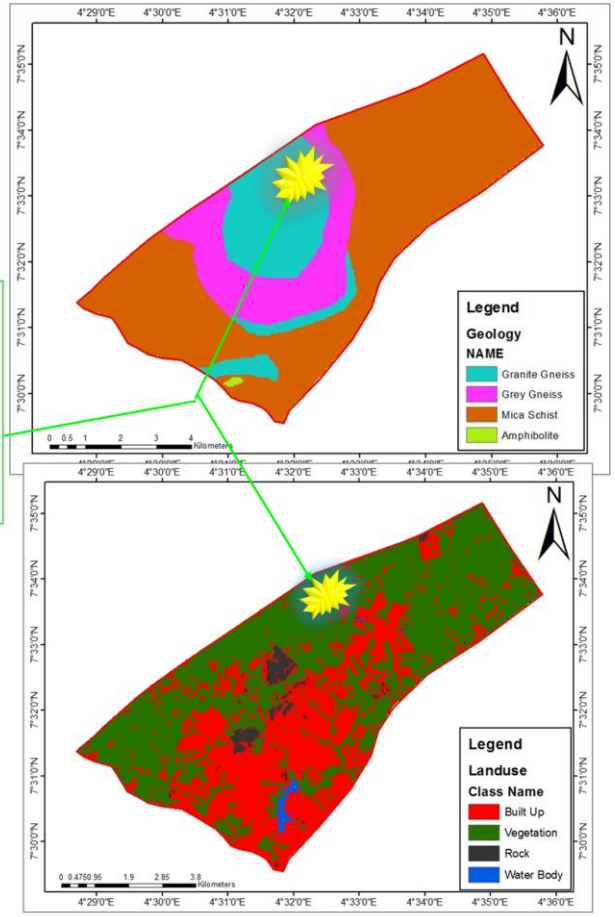
A multi-criteria GIS analysis integrating geology, land use, stream networks, terrain slope and road access to identify candidate areas.

Site selection from fuzzy logic



Possible site for an airport in OAU Campus, Nigeria

*The geology, land use, stream network, slope, and road network all favor the siting of an airport within Obafemi Awolowo University (OAU) campus, near latitude **7.53°N** and longitude **4.53°E**, and in its surrounding areas*

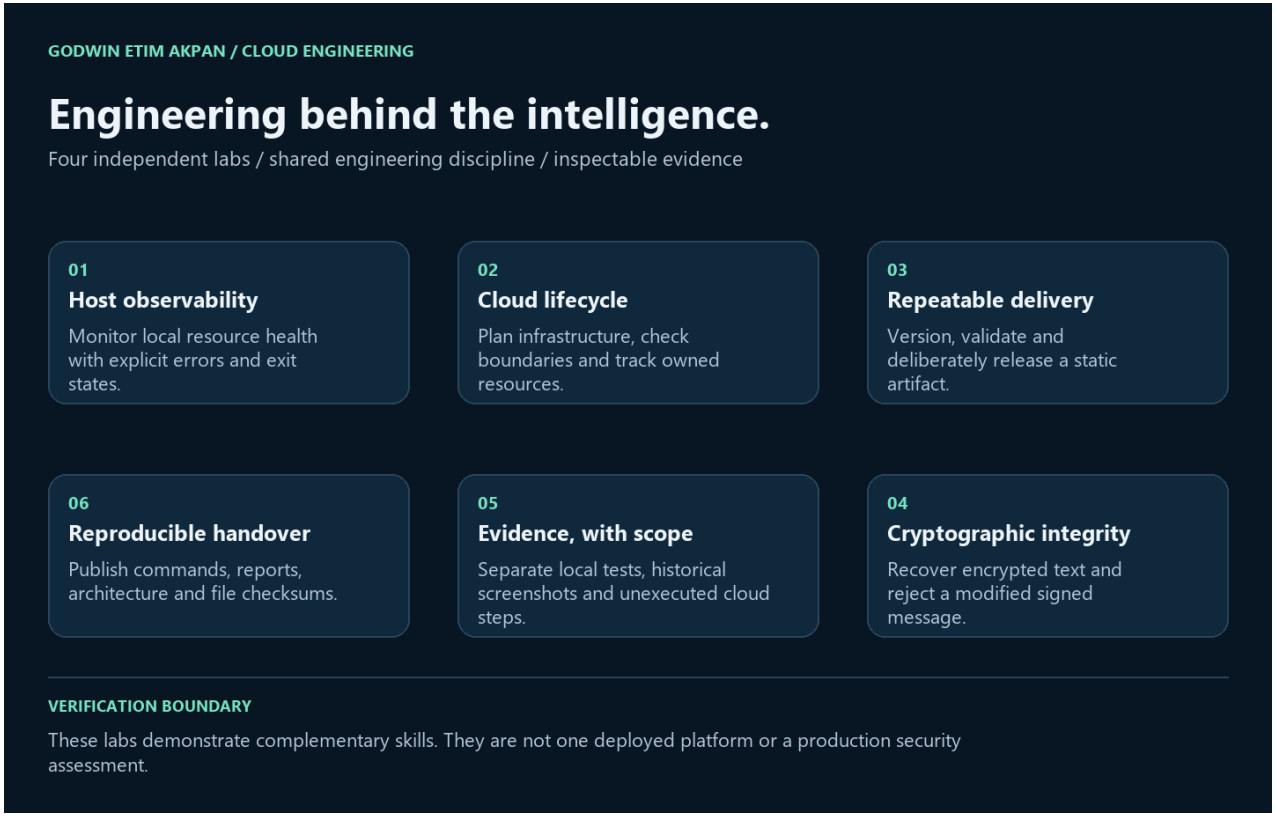


Fuzzy-logic suitability surface and candidate-site evaluation produced using ArcGIS, Landsat 8 and SRTM elevation data.

Methods demonstrated: Criteria standardisation · proximity analysis · slope and hydrology constraints · fuzzy overlay · candidate-site screening.

Monitoring cloud delivery and cryptographic integrity

Four practical labs with explicit distinctions between local checks, demonstrated functions and pending cloud integrations.



Lab	Verified evidence	Remaining integration
System health monitoring	11 tests; real local-host alert sample	SMTP delivery and Linux cron
AWS lifecycle	14 offline tests; ownership and cleanup logic	New AWS resource execution
Azure static delivery	5 artifact checks; supplied validation/deploy workflow	Azure deployment and endpoint check
Applied cryptography	4 real GnuPG checks; 5 offline failure tests	Key Vault remains a separate plan

Explore four architecture-led engineering case studies

Academic qualifications and professional development

Award status, professional certification, course completion and stackable recognition are distinguished explicitly.

MSc Big Data Technologies

University of East London / UNICAF · 2024–2026

All 180 credits completed and passed; final award pending.

Master of Technology in Space Science and Technology

Federal University of Technology, Akure / ARCSSTE-E · Awarded October 2018

Remote Sensing and Geoinformation Science · CGPA 4.50/5.00 · WES U.S. equivalency: master's degree.

BSc Chemistry

University of Uyo · 2010 · Second Class Honours Upper Division · WES U.S. equivalency: bachelor's degree.

Verify the WES evaluations

Focus	Selected credentials and development
Data and geospatial	IBM Data Science Professional Certificate V3 (October 2025); Esri Spatial Data Science (October 2024); Enterprise GIS MOOC (November 2026).
Cloud and automation	Google IT Automation with Python Professional Certificate (January 2026); Cisco Networking Essentials and NDG Linux Essentials course completion.
Security	CompTIA CySA+, Security+ and Splunk Core Certified User. CSAP is stackable recognition based on CySA+ and Security+. Per Scholas cybersecurity training is distinct from exam certification.
Public health	Liberia FETP Intermediate (December 2023) and Frontline (June 2022); Applied Epi R course (40 hours, August 2024); University of Washington health project management (7.5 CEUs, September 2024).

View credential badges and selected certificate evidence

Publications, Learning and Collaboration

Publications and research

I have authored and co-authored 10+ peer-reviewed publications covering infectious-disease surveillance, spatial epidemiology, GIS, GeoAI and public health intelligence.

Google Scholar

ORCID 0000-0001-8204-9219

ResearchGate

Current professional focus

- Reproducible data engineering and spatial QA
- Enterprise GIS and PostgreSQL/PostGIS workflows
- Cloud lifecycle controls, monitoring and delivery
- GeoAI validation, interpretability and spatial statistics
- Geospatial AI evaluation and analytical governance

OPEN TO OPPORTUNITIES

Data analytics · data engineering · enterprise GIS · GeoAI · spatial data science · machine learning · public health intelligence · secure decision-support systems

CONNECT

Portfolio

GitHub

LinkedIn

Email

Committed to building secure, explainable and practical data solutions that create measurable impact.